Dokumentasi Rumah Daun (Toko Tanaman Hias)

Tugas Personal Lab ke-1, Week 6 Topik: Membangun Toko Web Online (Frontend & Backend Development for Web)

A. Informasi Project

- **Judul**: Rumah Daun, Toko Tanaman Hias Online
- **Stack**:
  - Frontend: React 18 + Vite + React Router 6 + Axios
  - Backend: Node.js + Express 4 + Mongoose 8
  - Database: MongoDB Atlas (cloud)
- **Kategori Produk**: Indoor, Outdoor, Sukulen, Pot & Aksesoris

B. Persiapan

1. Install Node.js

Download dan install dari https://nodejs.org (versi LTS, 20.x atau lebih baru). Verifikasi:

```
node --version
npm --version
```

2. Setup MongoDB Atlas (Gratis)

1. Buka https://www.mongodb.com/cloud/atlas/register dan buat akun.
2. Buat **New Project**, namanya bebas, mis. `TokoTanaman`.
3. **Build a Database** → pilih tier **M0 FREE** → pilih region terdekat (mis. Singapore) → Create.
4. Pada **Security Quickstart**:
   - Username & password → simpan (mis. `admin` / `passwordkuat123`).
   - **Network Access** → Add IP Address → **Allow Access from Anywhere** (`0.0.0.0/0`) untuk kemudahan pengembangan.
5. Setelah cluster siap → klik **Connect** → **Drivers** → **Node.js** → copy **connection string**. Contoh:

   ```
   mongodb+srv://admin:<password>@cluster0.abcde.mongodb.net/?retryWrites=true&w=majority
   ```

   Ganti `<password>` dengan password asli, dan tambahkan nama database sebelum tanda tanya:

   ```
   mongodb+srv://admin:passwordkuat123@cluster0.abcde.mongodb.net/tokoTanaman?retryWrites=true&w=majority
   ```

C. Menjalankan Backend

1. Masuk ke folder backend:

   ```
   cd backend
   ```

2. Install dependencies:

   ```
   npm install
   ```

3. Buat file `.env` (copy dari `.env.example`) dan isi `MONGO_URI` dengan connection string dari Atlas:

   ```
   PORT=5000
   MONGO_URI=mongodb+srv://admin:passwordkuat123@cluster0.abcde.mongodb.net/tokoTanaman?retryWrites=true&w=majority
   ```

4. Isi data contoh (opsional, 8 produk tanaman):

   ```
   npm run seed
   ```

5. Jalankan server:

   ```
   npm run dev
   ```

   Server jalan di **http://localhost:5000**. Buka link tersebut, harusnya muncul JSON info endpoint.

D. Menjalankan Frontend

Buka **terminal baru** (biarkan backend tetap jalan).

1. Masuk ke folder frontend:

   ```
   cd frontend
   ```

2. Install dependencies:

   ```
   npm install
   ```

3. (Opsional) buat `.env` jika backend bukan di localhost:5000. Default sudah cukup.

4. Jalankan dev server:

   ```
   npm run dev
   ```

   Browser otomatis terbuka di **http://localhost:5173**.

E. Menguji API dengan Postman/cURL

Get all products

```
curl http://localhost:5000/api/products
```

Get by kategori

```
curl "http://localhost:5000/api/products?category=Sukulen"
```

Create product

```
curl -X POST http://localhost:5000/api/products \
  -H "Content-Type: application/json" \
  -d '{"name":"Lidah
Mertua","price":45000,"stock":20,"category":"Indoor","description":"Sansevier
ia","image":"https://..."}'
```

Update product

```
curl -X PUT http://localhost:5000/api/products/<id> \
  -H "Content-Type: application/json" \
  -d '{"price":50000}'
```

Delete product

```
curl -X DELETE http://localhost:5000/api/products/<id>
```

F. Struktur Routing Frontend

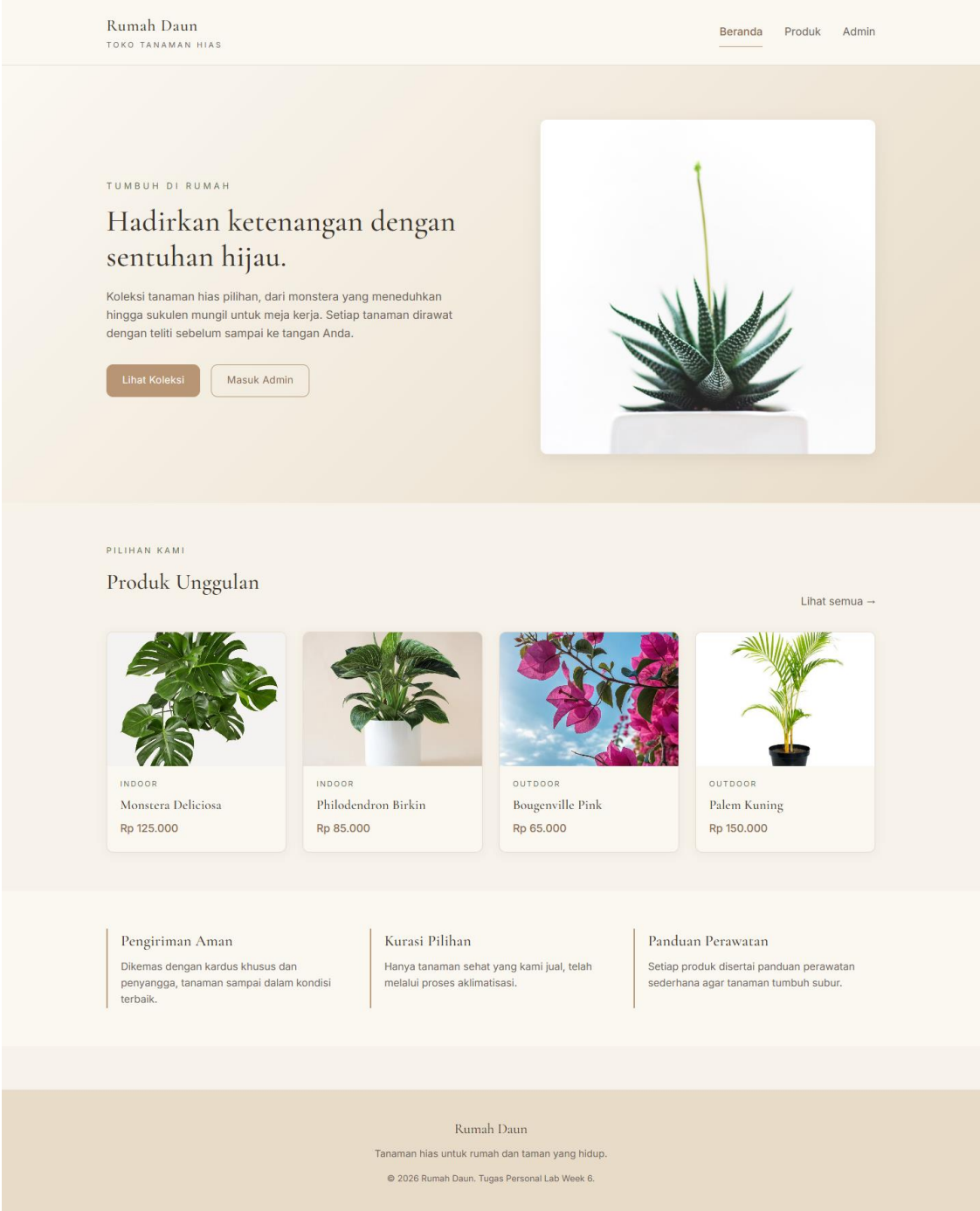
URL	Halaman	Keterangan
`/`	Home	Landing + featured produk
`/products`	Products	Grid + filter + search
`/products/:id`	ProductDetail	Detail satu produk
`/admin`	Admin	Tabel produk + tombol CRUD
`/admin/new`	AdminForm	Form tambah produk
`/admin/edit/:id`	AdminForm	Form edit produk

G. Desain & UX

- **Palette earthy tones**: krem (`#f7f2ea`), clay (`#b89574`), moss (`#8a9a7b`), terracotta (`#c08464`), ink (`#3a332b`).
- **Tipografi**: Cormorant Garamond (serif, untuk heading) + Inter (sans-serif, untuk body).
- **Layout responsif** dengan CSS Grid & Flexbox.
- **Tanpa framework CSS**, styling ditulis langsung supaya ringan dan mudah dikustomisasi.

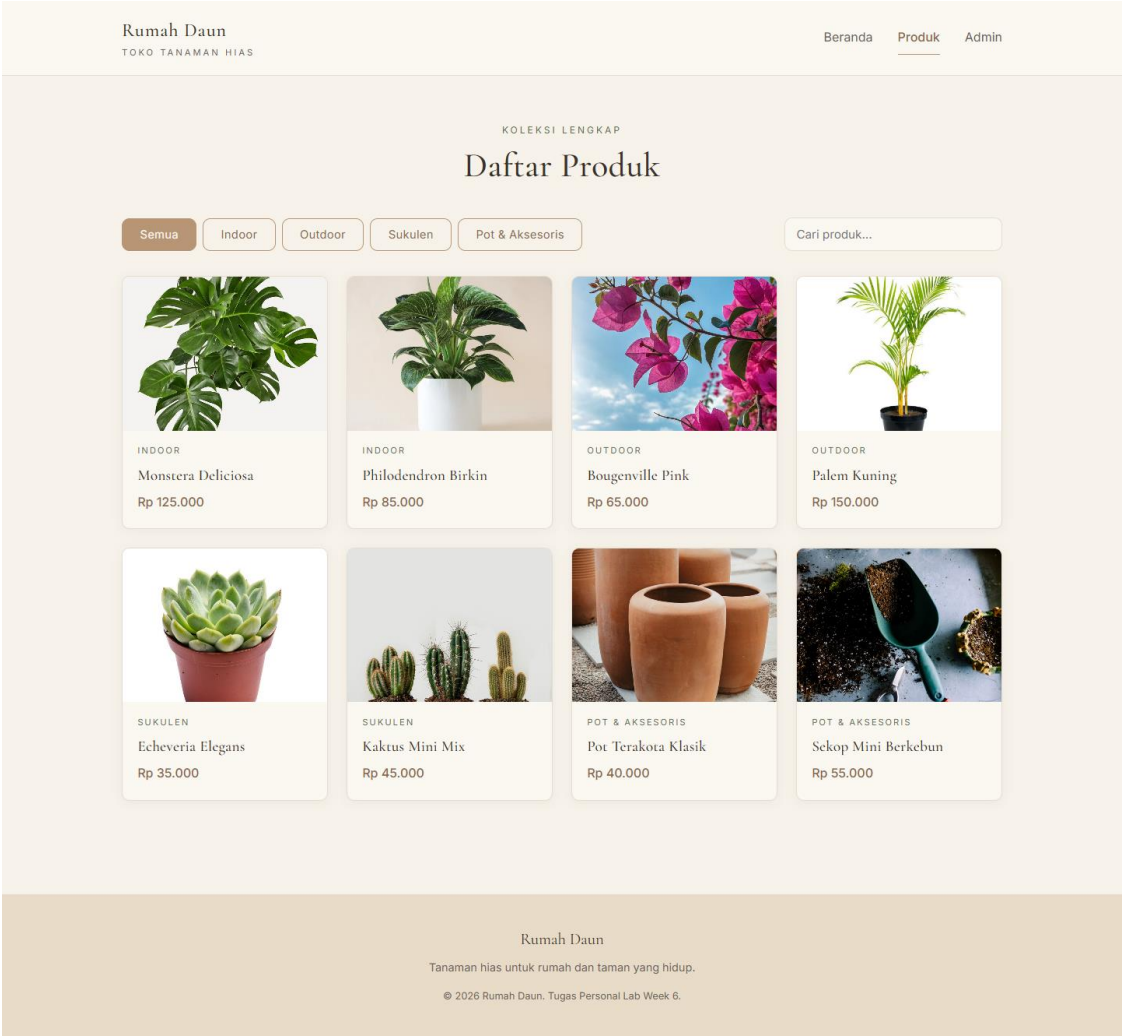
H. Screenshot Hasil Aplikasi

1. Halaman Beranda



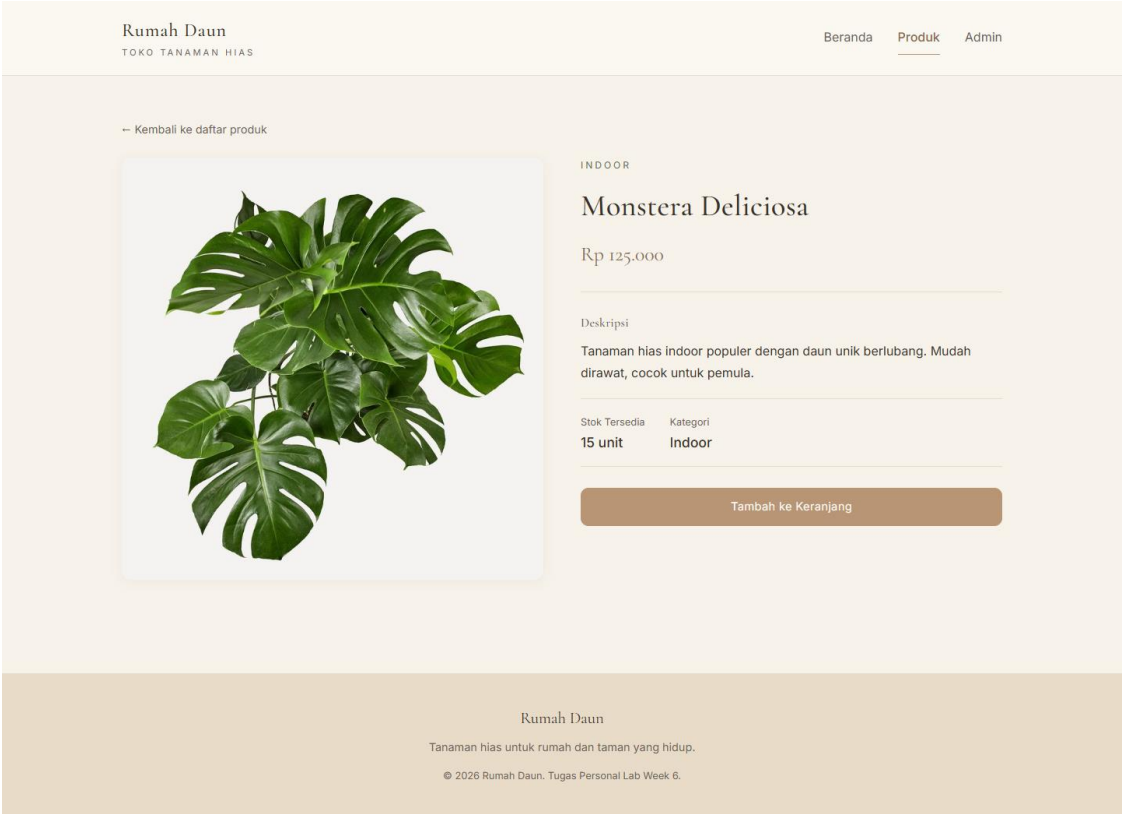
Halaman Beranda

2. Daftar Produk



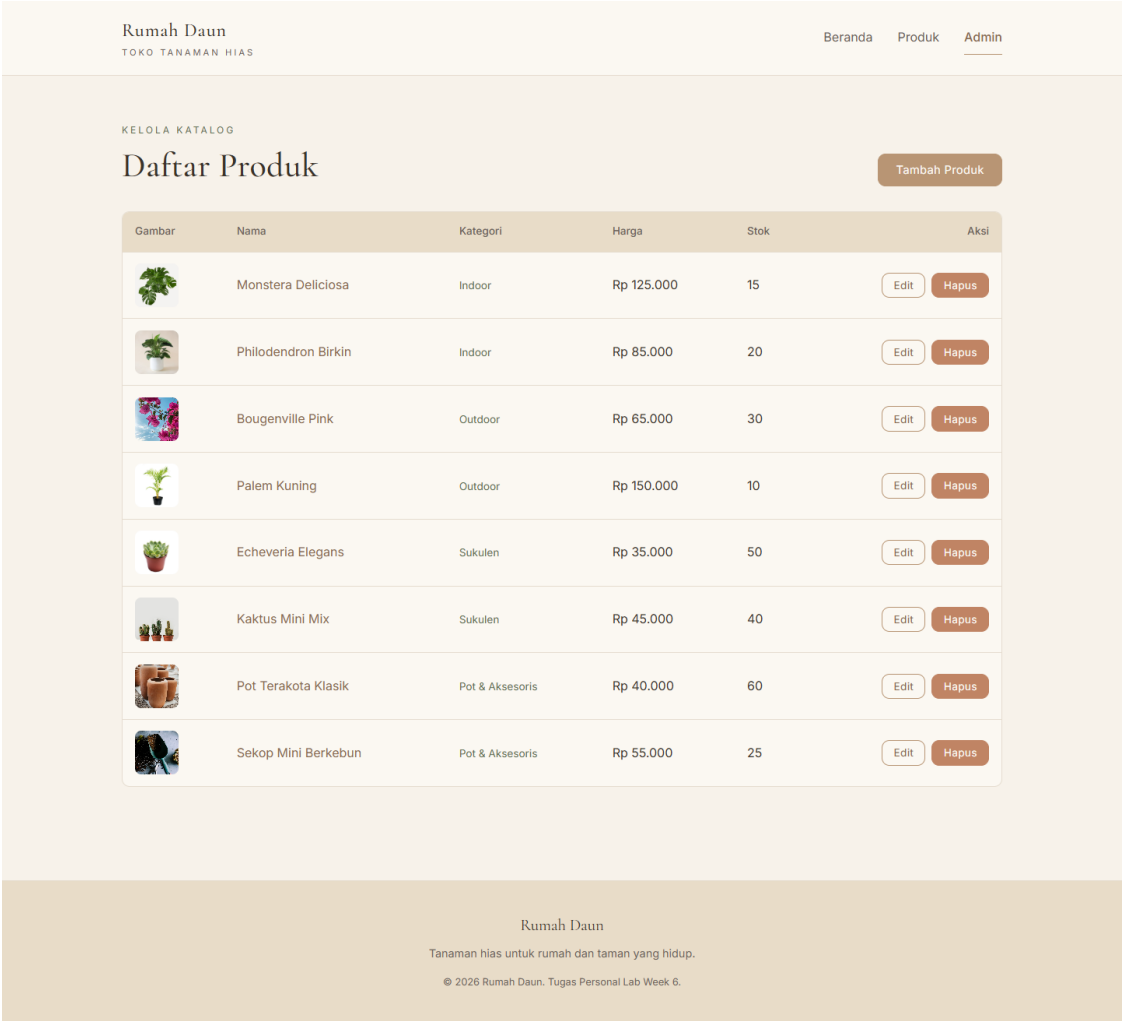
Daftar Produk

3. Detail Produk



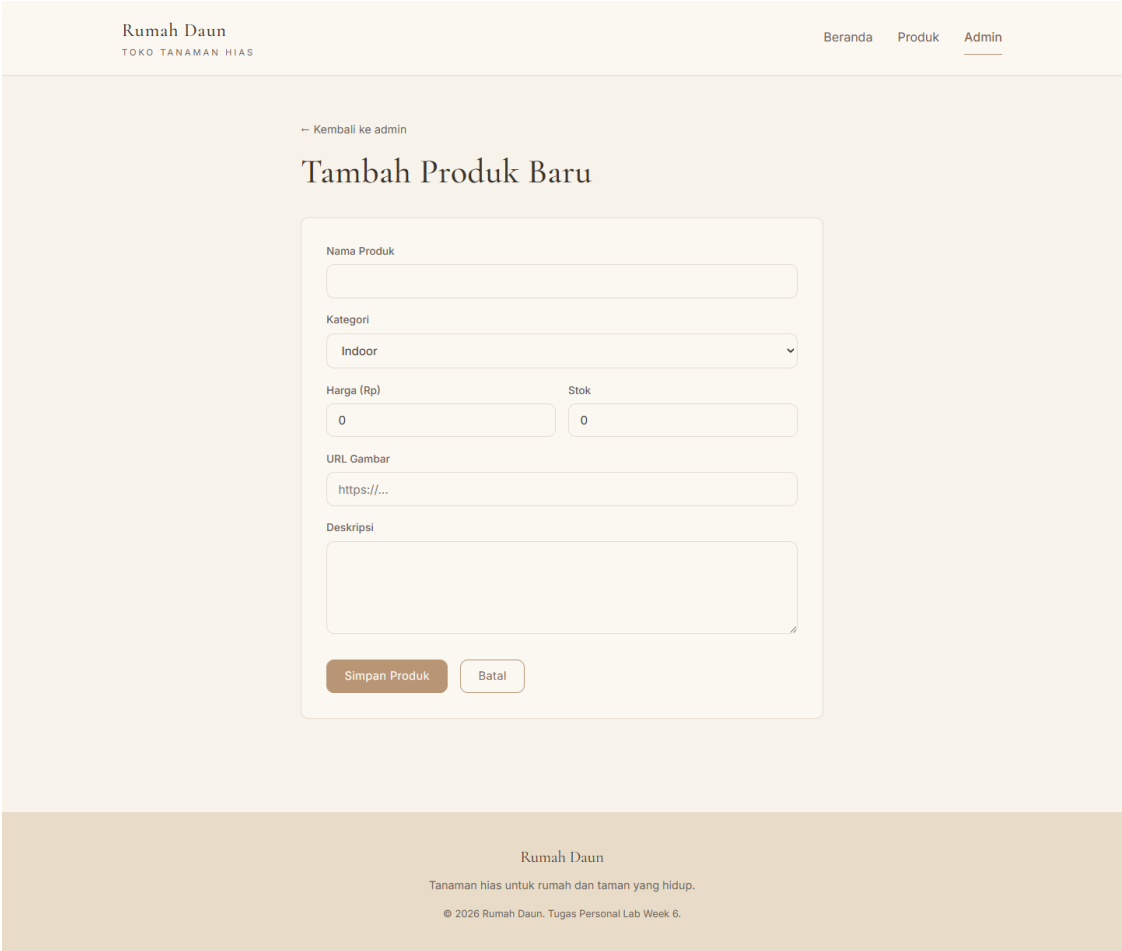
Detail Produk

4. Admin (Daftar Produk)



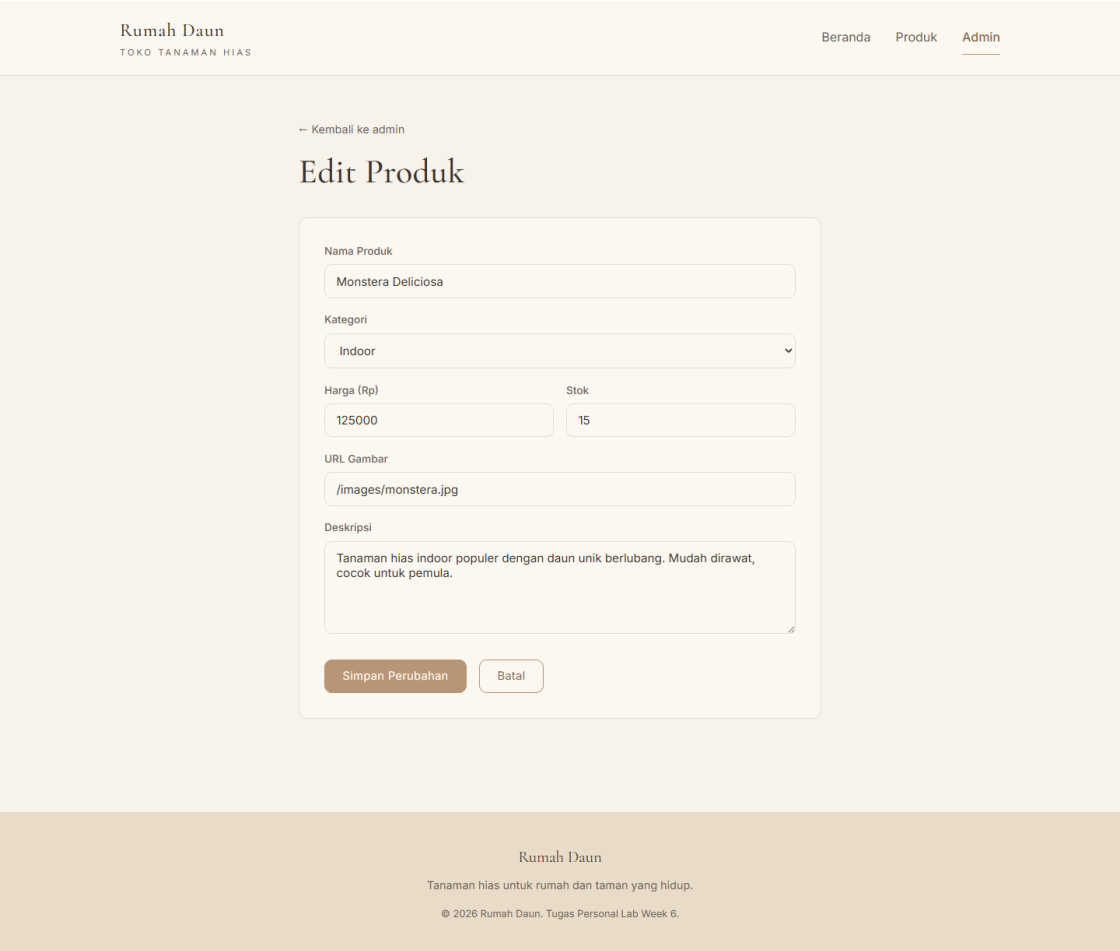
Admin List

5. Admin (Tambah Produk)



Admin Form Tambah

6. Admin (Edit Produk)



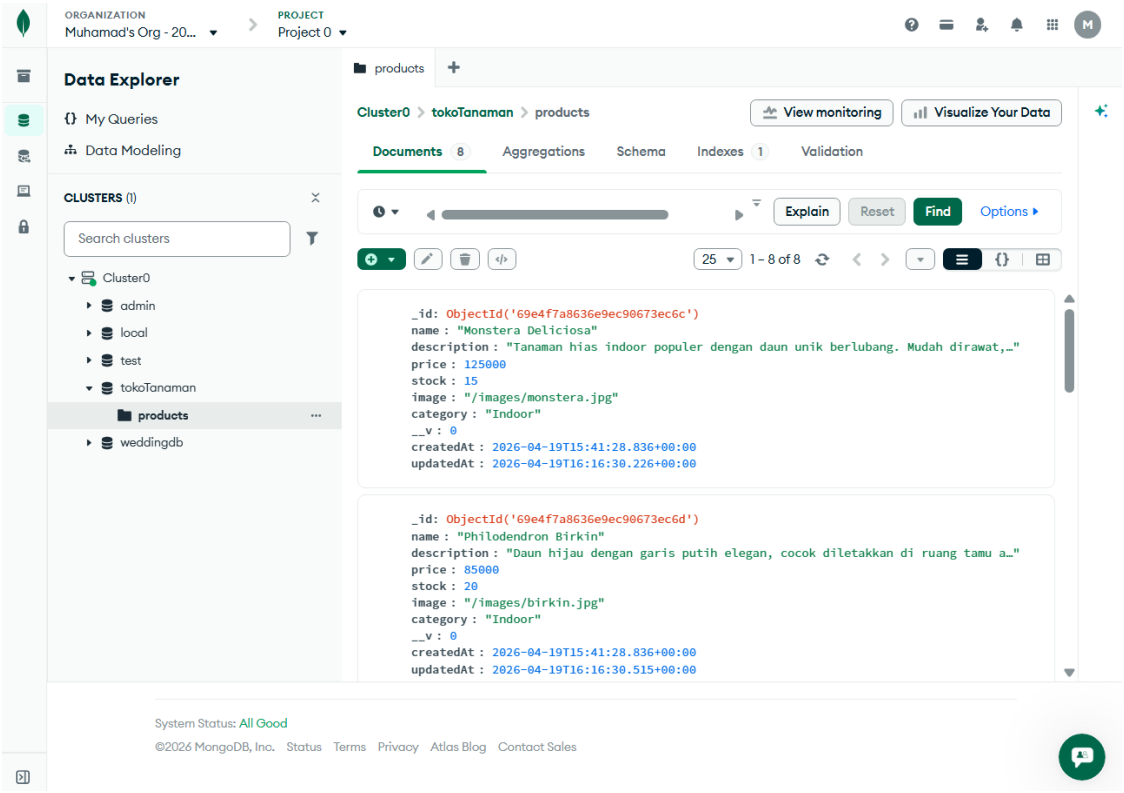
Admin Form Edit

7. Terminal Backend

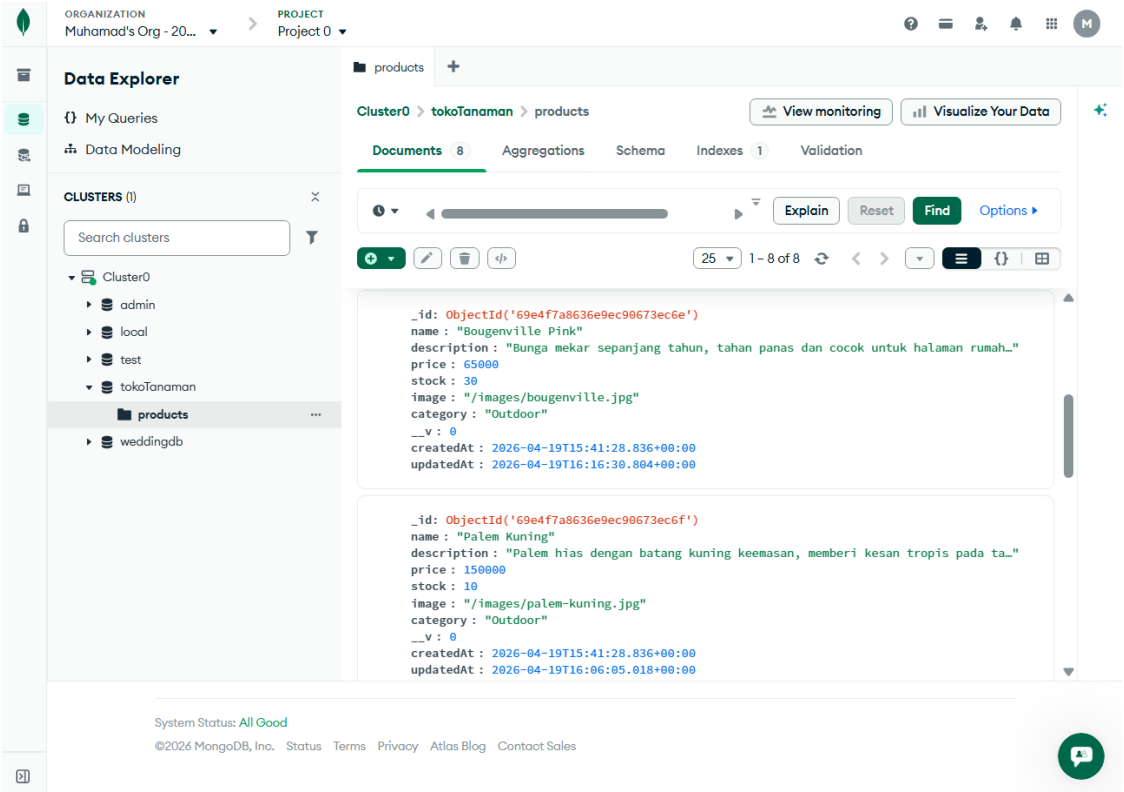
Terminal Backend

8. MongoDB Atlas Collection

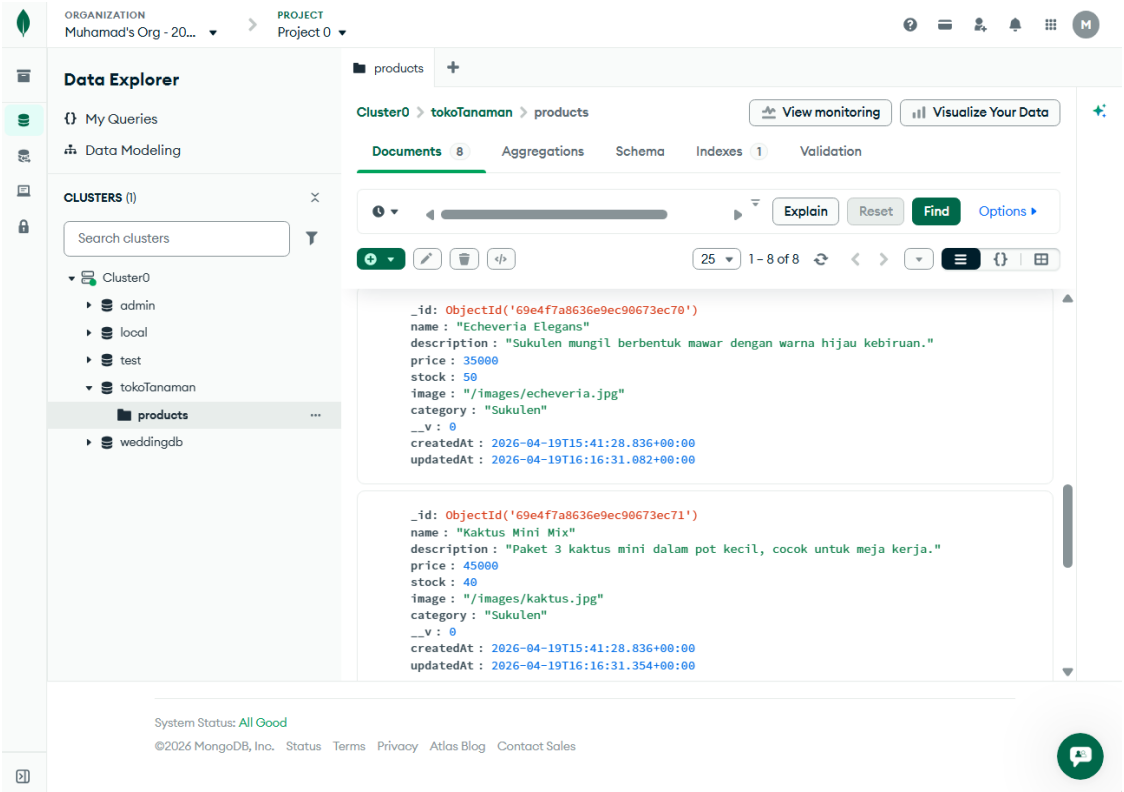
Data produk tersimpan di cluster `Cluster0` → database `tokoTanaman` → collection `products` dengan total 8 dokumen.



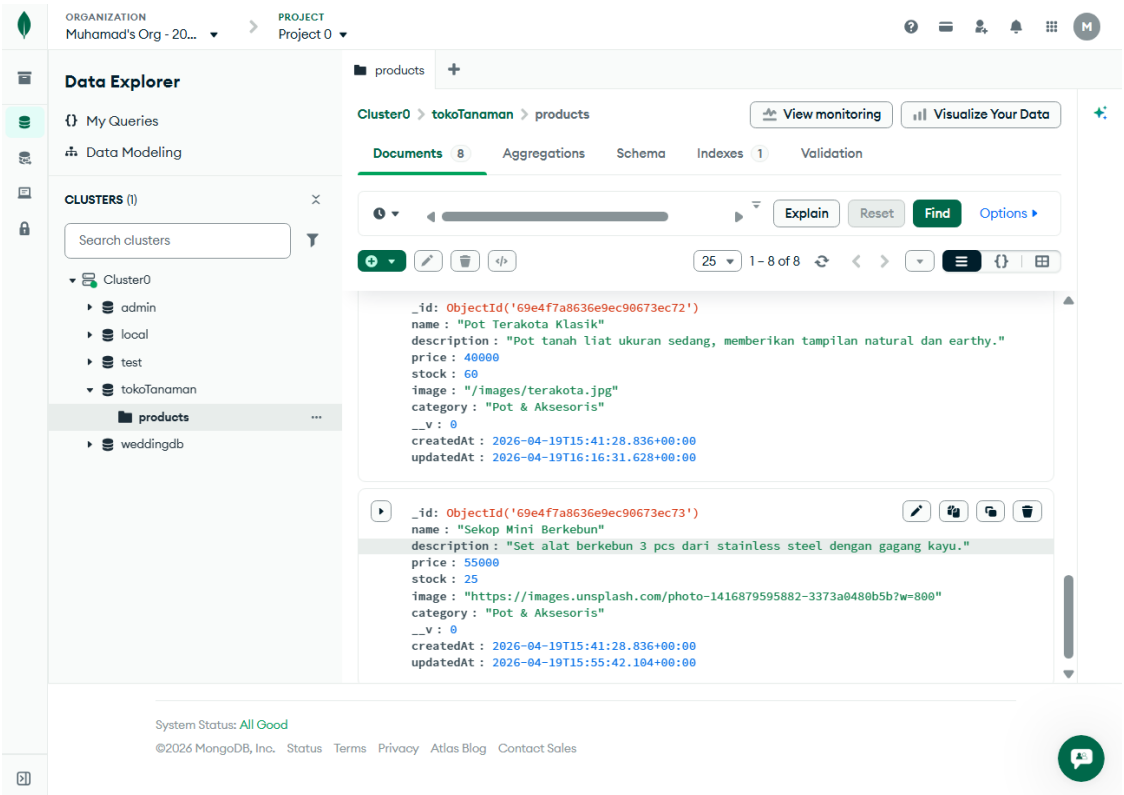
Atlas Data Explorer 1



Atlas Data Explorer 2



Atlas Data Explorer 3



Atlas Data Explorer 4

I. Troubleshooting

- **Backend error `MongoDB connection error`** → Cek `MONGO_URI`, pastikan IP sudah di-whitelist di Atlas (Network Access).
- **Frontend error `Network Error` / CORS** → Pastikan backend jalan di port 5000. Kalau port berbeda, edit `frontend/.env`: `VITE_API_URL=http://localhost:PORT/api`.
- **Produk tidak muncul** → Jalankan `npm run seed` di backend.